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$$(x_n)_n = (0.1, 1.1, 2.1, 0.01, 1.01, 2.01, 0.001, 1.001, 2.001, \dots)$$

We bekijken de drie deelrijen:

$$x_{3k+1} = 10^{-k}$$

$$x_{3k+2} = 1 + 10^{-k}$$

$$x_{3k+3} = 2 + 10^{-k}$$

Dus:

$$\lim_{k \rightarrow +\infty} x_{3k+1} = 0$$

$$\lim_{k \rightarrow +\infty} x_{3k+2} = 1$$

$$\lim_{k \rightarrow +\infty} x_{3k+3} = 2$$

Daarom bestaat $\lim_{n \rightarrow +\infty} x_n$ niet.

$$\text{adh}_{n \rightarrow +\infty} x_n = \{0, 1, 2\}$$

$$\liminf_{n \rightarrow +\infty} x_n = 0$$

$$\limsup_{n \rightarrow +\infty} x_n = 2$$

$$\begin{aligned} \lim_{n \rightarrow +\infty} x_n &\text{ bestaat niet} \\ \text{adh}_{n \rightarrow +\infty} x_n &= \{0, 1, 2\} \\ \liminf_{n \rightarrow +\infty} x_n &= 0 \\ \limsup_{n \rightarrow +\infty} x_n &= 2 \end{aligned}$$

References